Examiner only

8. *(a)* Copper can be extracted from the mineral chalcopyrite. The mineral is smelted by heating with air. The equation for this reaction is as follows.

$$4CuFeS_2 + \ldots\ldots\ O_2 \longrightarrow \ldots\ldots\ Cu + \ldots\ldots\ FeO + \ldots\ldots\ Fe_2O_3 + \ldots\ldots\ SO_2$$

(i) Balance the equation above. [1]

(ii) Give the full electronic configuration of a copper atom. [1]

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(iii) A sample of rock contains 1.30 % by mass of chalcopyrite. Assuming this is the only source of copper in the rock, calculate the percentage by mass of copper in the sample. [2]

Percentage = ... %

(iv) During the smelting Cu_2S is produced. This is collected and blown with air to produce copper.

$$Cu_2S + O_2 \longrightarrow 2Cu + SO_2$$

This is classified as a redox reaction. Use oxidation numbers to explain which elements have been oxidised and which reduced. [3]

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(b) Sulfur dioxide can react with oxygen and water to form sulfuric acid. Although sulfuric acid is a strong acid, it does not have to be a concentrated acid.

Explain the difference between the terms *strong acid* and *concentrated acid.* [2]

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(c) (i) A student was given an aqueous solution of an acid HX and was asked to find out if it was a strong or weak acid. 25.0 cm^3 of the acid HX required 15.90 cm^3 of 0.0125 mol dm^{-3} sodium hydroxide solution for complete neutralisation.

Calculate the concentration of the acid. Assume that HX reacts with the sodium hydroxide in a 1:1 molar ratio. [2]

Concentration = ... mol dm^{-3}

(ii) A teacher measured the pH of the aqueous solution and found it to be 2.10. She told the student that the acid in the solution must be a strong one. Is she correct? Justify your answer. [2]

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Examiner only

(d) The commonest complex ion that copper forms in solution is the hexaaquacopper(II) ion, $[Cu(H_2O)_6]^{2+}$. If concentrated hydrochloric acid is added to a solution containing hexaaquacopper(II) ions a new complex forms as the following equilibrium is established.

$$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 6H_2O(l)$$

blue yellow-green

The forward reaction is endothermic.

(i) State and explain what you would observe on adding water to the equilibrium mixture. [2]

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(ii) State and explain what you would observe on heating the equilibrium mixture. [2]

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